



What Gen2 Can Do For You

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Agenda

- **Gen2 is here, now**

- And it works

What should I look for?

- **A Gen2 tag is not a bar code**

- Allows inventory, read, write, lock, kill

What is the impact on my operations?

- **Gen2 has a bright future**

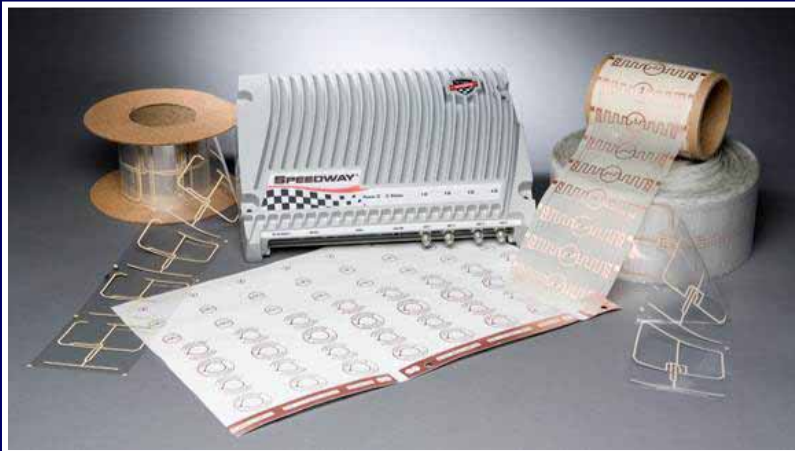
- Tagging items using UHF Gen2

Can I use Gen2 for pallets and for items?

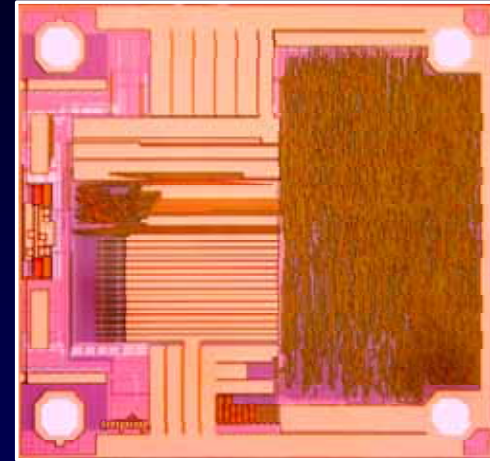
Gen2 is Here, Now

- **Gen2 Monza™ silicon first introduced in April, 2005**
 - 96-bit EPC
 - 10m read, 8m write range
 - Writes 800 tags/minute
 - 100 million Monza™ tags shipped
- **GrandPrix™ system solution**
 - Reader, silicon, tags

GrandPrix™ System Solution



Impinj Monza™ Gen2



Speedway™ Gen2 Reader



What You Should Look For

1. Readers certified for dense-reader mode

- Allows many readers to operate simultaneously

2. Tags with high sensitivity

- Ensures that tags *inside* a pallet get read

3. Tags with worldwide operation

- Ensures that tags can be read anywhere

4. Tags that reject interference

- Critical to dense-reader operation

5. Readers without ghost reads

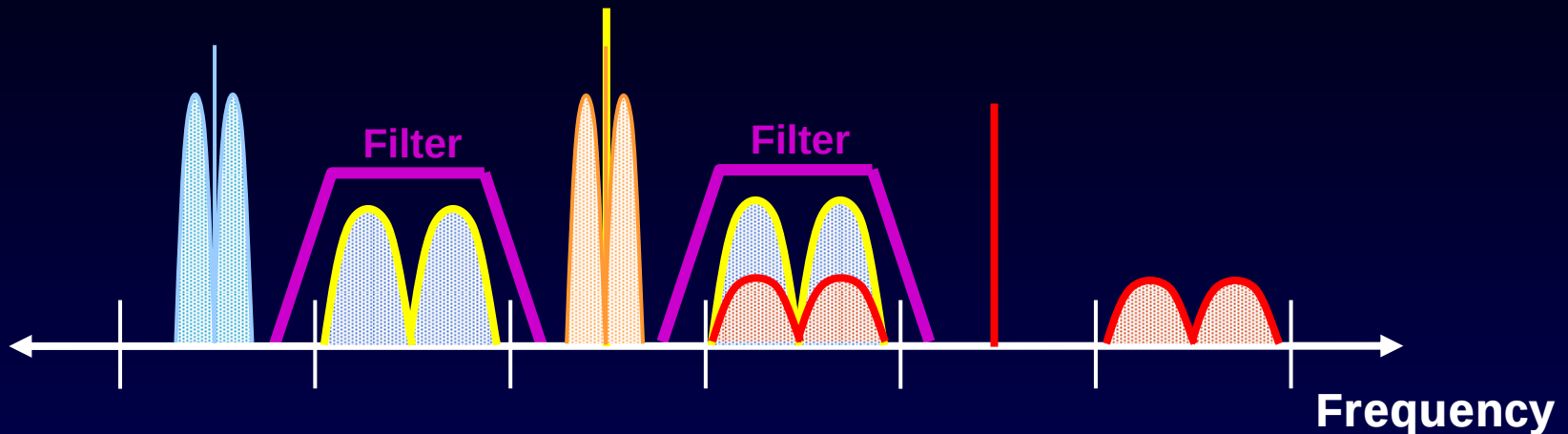
- < 1 ghost read / year

Dense-Reader Goal

- **10's or hundreds of readers operating simultaneously**
 - Reliably reading pallets & items
 - Without needing synchronization



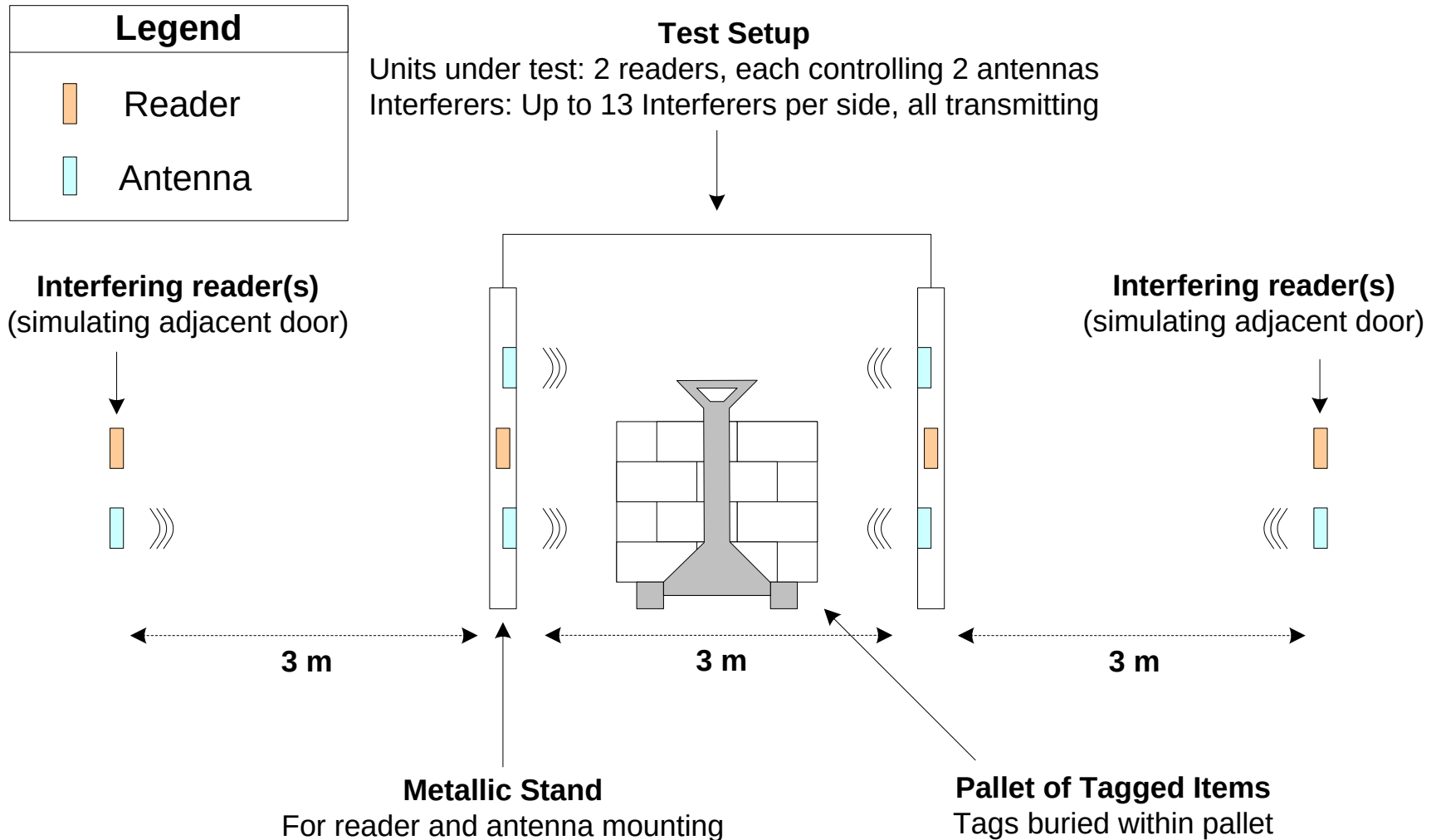
1. Dense-Reader Mode



Readers collide with readers but not tags
Readers filter interfering readers from their tag responses

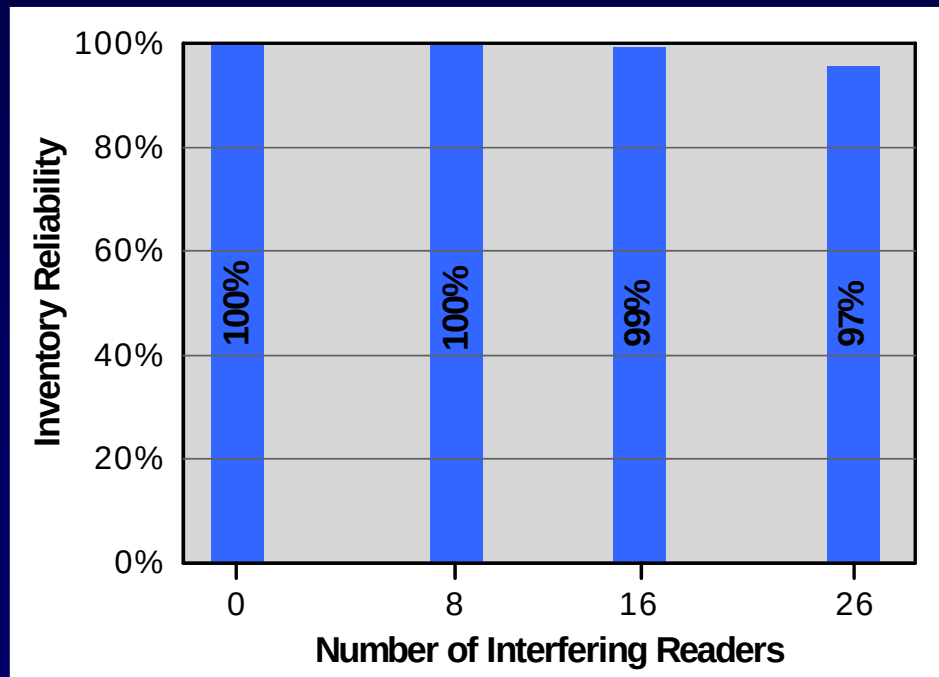
- **Separates tags and readers in frequency**
 - Prevents reader—tag interference
- **Eliminates need for reader synchronization**
 - Eliminates poor-performing listen-before-talk
- **Defined algorithmically**
 - Different frequency plans for different regulatory regions

Dense-Reader Mode Testing



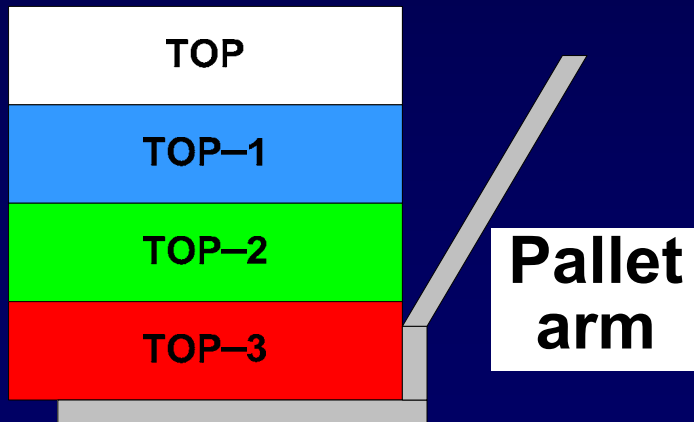
Dense-Reader Mode Works

- Impinj Speedway™ reader on either side of dock door (FCC certified)
- Avery AD-620 (Triflex™) tags on each of 40 boxes of Caress® soap
- All readers transmitting simultaneously
- **Inventory reliability is achievable**



2. Tags With High Sensitivity

- **Red tags are the hardest to read**
 - Buried at bottom of pallet
 - Near the metal arm
- **High sensitivity = long read range = higher reliability**
 - Need tags buried deep inside pallet to turn on



TOP		Pallet arm	
x-	Tag 2	x-	x-
Tag 31	Tag 15	Tag 36	Tag 9
x-	x-	x-	x-
Tag 34	Tag 35	Tag 28	Tag 1
	x-	Tag 23	x-

TOP-1		Pallet arm	
Tag 38	Tag 20	x-	Tag 19
x-	x-	Tag 17	Tag 6
Tag 5	Tag 29	x-	x-
x-	x-	Tag 26	Tag 37
x-	Tag 24		

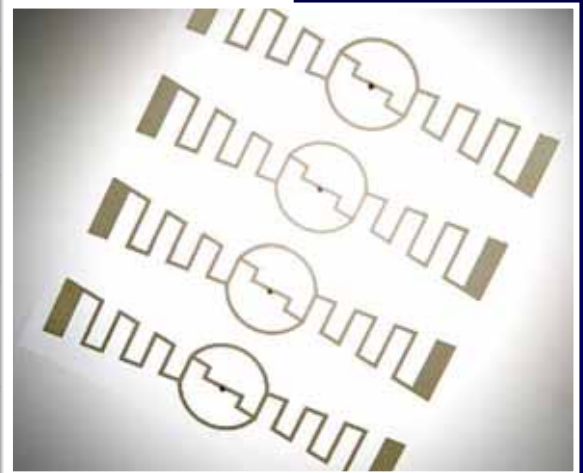
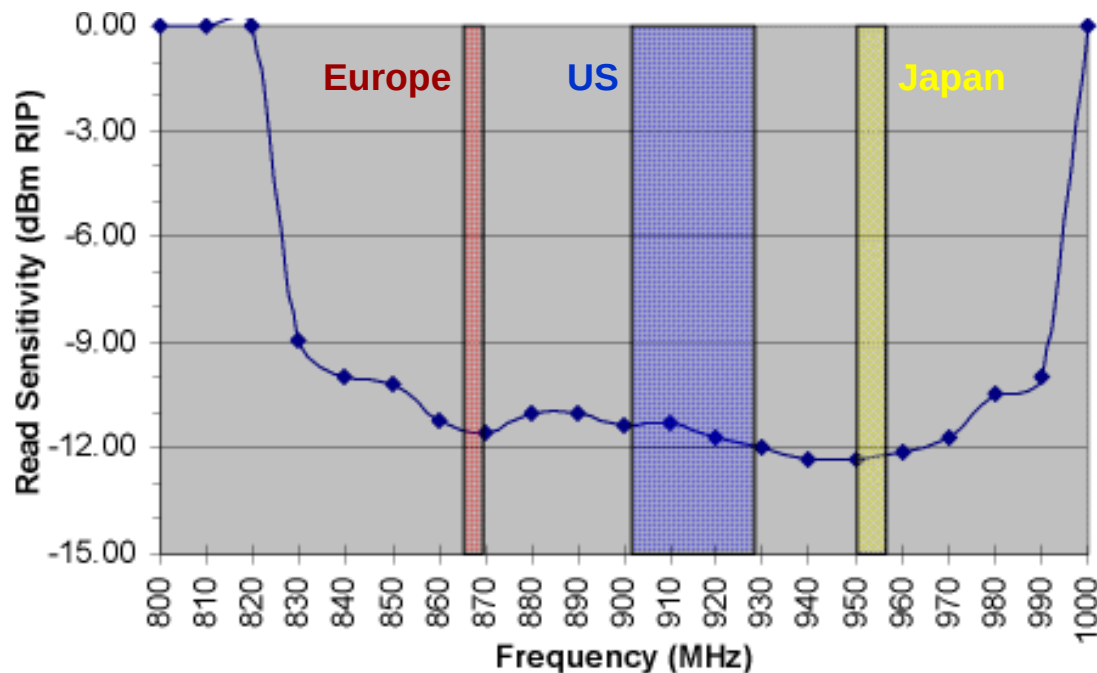
TOP-2		Pallet arm	
Tag 33	x-	x-	x-
x-	Tag 11	Tag 3	Tag 4
Tag 18	x-		x-
Tag 27	Tag 8	Tag 14	Tag 10
x-		Tag 25	x-

TOP-3		Pallet arm	
Tag 21	Tag 16	x-	Tag 12
x-	x-	Tag 32	Tag 22
Tag 13	Tag 30	x-	x-
x-		Tag 39	Tag 7
x-	Tag 40		x-

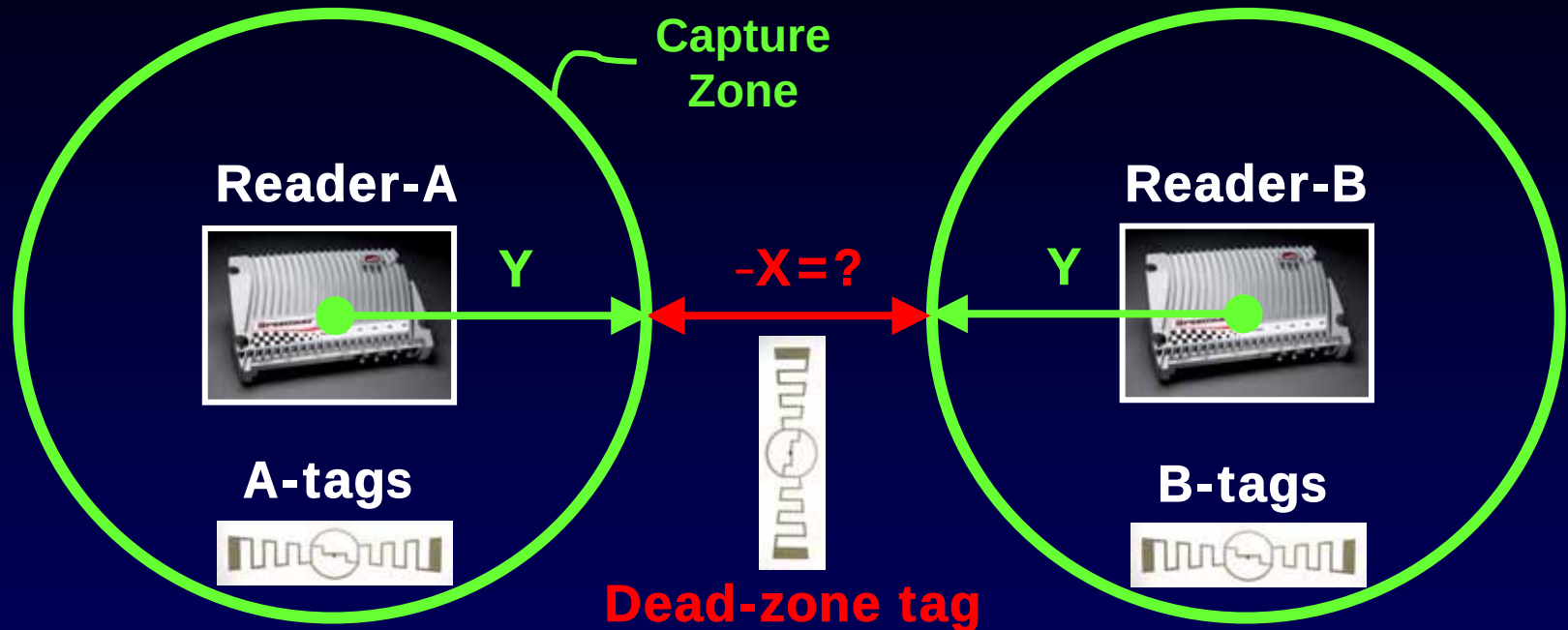
3. Worldwide Tag Operation

- **Gen2 tags can operate worldwide**
 - Same tag operates in Asia/Europe/US
 - Have exceptional sensitivity across all regions
 - No need to use different tags for different locations

Propeller Tag Frequency Response (Monza™ Silicon)



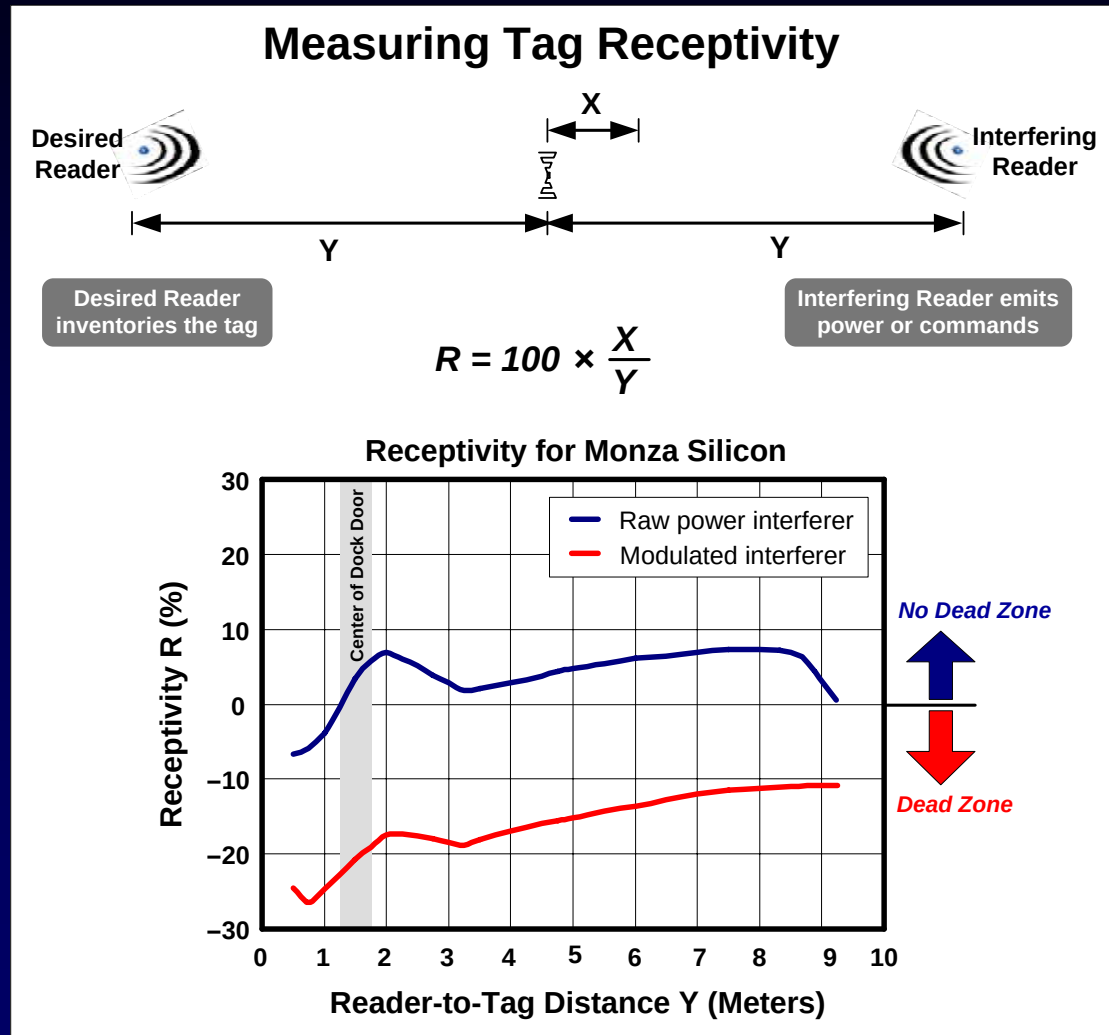
4. Tags Must Reject Interference



- Reader signals interfere at the tag
- Tags in “**dead zone**” between “**capture zones**” cannot differentiate reader-A from reader-B
- **Question: How big is the dead zone?**

Tags Can Reject Interference

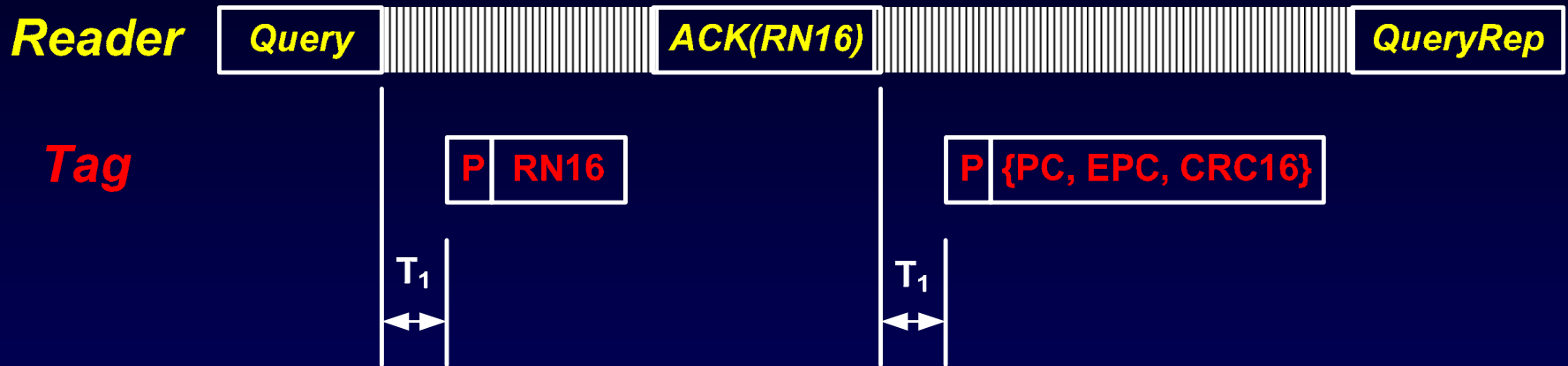
- **R shows reader-field overlap for which the tags will still function**
 - +10 = 10% overlap
 - -10 = 10% dead zone
- **Good tag has $R > 0$**
 - No dead zone
- **Plot shows that Monza interference rejection allows simultaneous transmitting readers**
 - Dock door has no dead zone where tags cannot be read



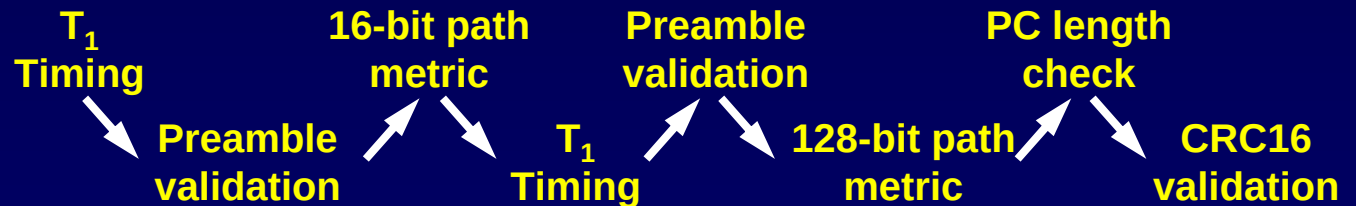
1 MHz interferer frequency offset

5. No Ghost Reads

- Probability of a ghost read in Gen2 is $\ll 1/\text{year}$



Tags must pass 8 checks to be "valid"



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- **Gen2 has a bright future**

- Tagging items using UHF Gen2

Can I use Gen2 for pallets and for items?

Fundamental Gen2 Operations

Operation	Function	Data Flow
<i>Inventory</i>	Singulate tags and receive their EPCs	Reader to network
<i>Read</i>	Read tag memory	Bidirectional
<i>Write</i>	Write tag memory	Bidirectional
<i>Lock</i>	Permalock, lock, or unlock tag memory	Bidirectional
<i>Kill</i>	Render a tag permanently inoperative	Bidirectional

The Lasting Benefit of Gen2

- **Read:** Read user memory for additional item information
- **Write:** Write travel history and destination into a baggage tag
- **Lock:** Store an in-service date into a tag and lock it
- **Kill:** Disable a tag prior to consumer checkout

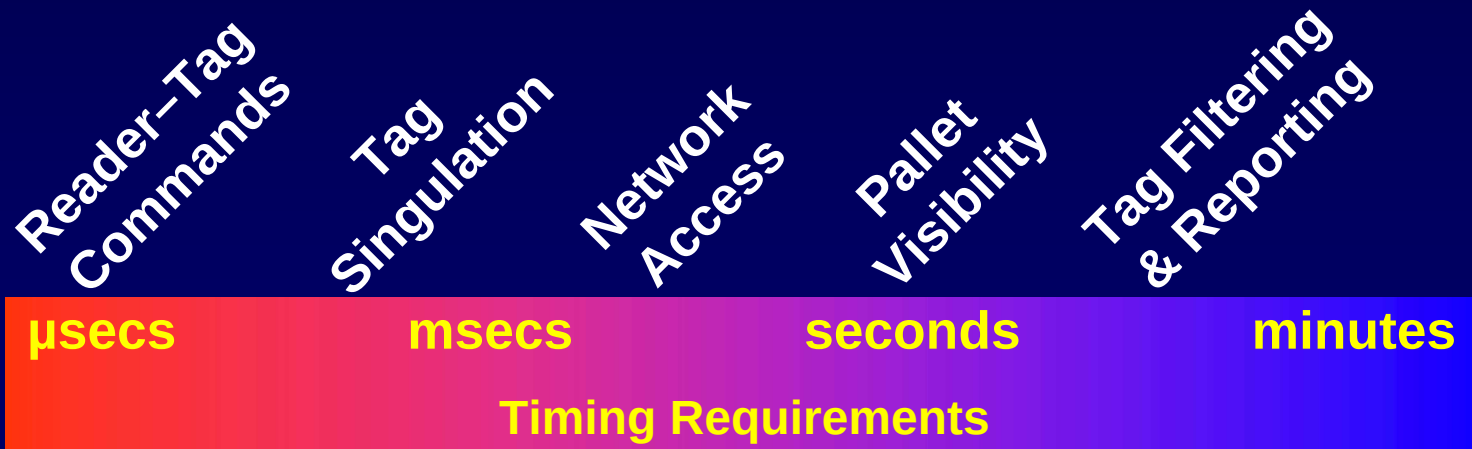


Gen2 tags are not just bar-code replacements



The Technical Issue

- **Read, write, lock, and kill are linked to a tag's EPC**
 - What you write
 - Which password to use
- **Bidirectional network access takes too long**
 - “holding” a tag means a reader can't inventory another
 - A “released” tag is difficult to find again
- **Readers need to cache data locally**
 - Need to know what tags they expect to see



Impact on My Operations

- **Gen2 tags are remote data objects**
 - Enable functionality not possible with bar codes
 - Change the rules of inventory management
- **Gen2 readers must be “locally smart”**
 - Need prior knowledge of what tags they will see
 - Cached tag data
 - Password association table
 - Need to employ business rule-sets for read/write/lock/kill
- **End users that develop an RFID model based around tags as data objects, smart readers, and business rule-sets will enjoy all the power of Gen2 RFID**

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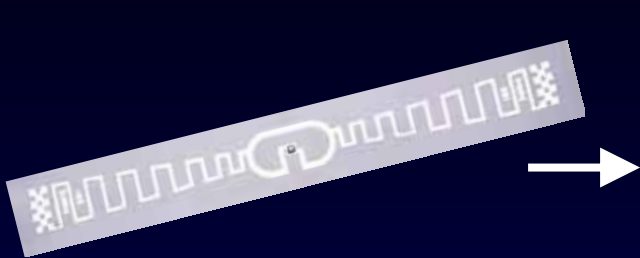
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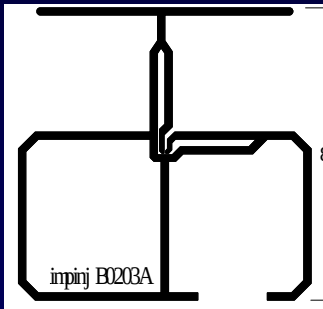
Supply Chain Fundamentals

- **Today's supply chain is a global, shared ecosystem**
 - No clear line between industry verticals
 - The definition of “case” and “item” varies by product
 - For some, 15% of merchandise is case-pack one-of-one
- **The industry needs a single RFID infrastructure**
 - Case and pallet tagging
 - Case-pack one-of-one tagging (printers, computers, etc)
 - Item tagging
- **Goal: A low-cost, global solution**
 - A single RFID protocol that can be used for any object anywhere in the world

Conventional Wisdom



“UHF doesn’t work on liquids”



“UHF tags don’t fit”

“UHF can’t work, it’s the Physics!”



“UHF doesn’t work on metals”

“You can’t stack UHF tags”

**The Conventional Wisdom
is Just Plain Wrong**

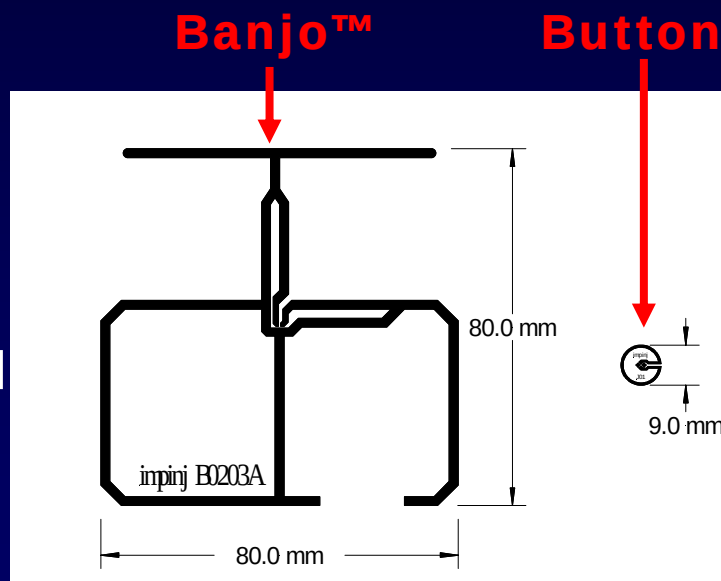
Near-Field versus Far-Field

Near-Field UHF RFID	Far-Field UHF RFID
Uses magnetic or electric fields • Inductive or capacitive coupling	Uses electromagnetic waves • Radiated energy
Short range • Ideal range < 1 wavelength	Long range • Ideal range > 1 wavelength
Rapid field attenuation	Gradual field attenuation
Unaffected by nearby dielectrics	Attenuated by nearby dielectrics

- **The wavelength**
 - 900 MHz UHF: 30 cm
- **Key takeaway**
 - UHF RFID can work in far-field and in near-field
 - All you need to do is change the reader and tag antennas

Gen2 Tags Can be Quite Small

- **Fact #1: UHF pallet tags need long range**
 - Require large far-field antennas
- **Fact #2: UHF item-level tags need short range**
 - Use small near-field antennas



Gen2 Works Fine on Liquids

- **Near-field UHF is not affected by liquids**
 - UHF Gen2 tags not only work on water, they work under water!



Gen2 Works Fine on Metals

- **All fields are affected by metals**
 - Near field and far field
- **UHF can take advantage of the metal**
 - Can tune the antenna for metal
 - UHF tags can work better on metal than in free space



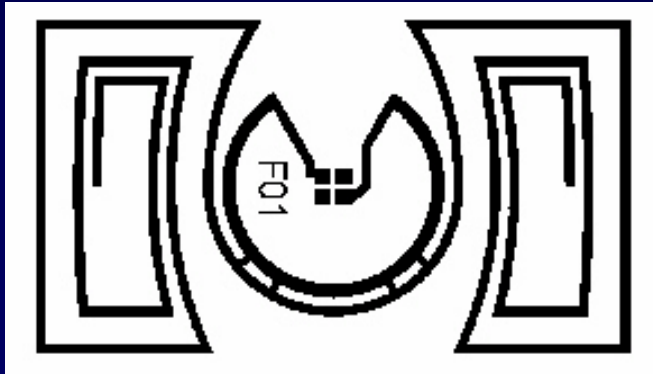
Gen2 Works Fine in Close Proximity

- **UHF enables simple loop-antenna designs**
 - Reduced tag-to-tag shielding
 - Stacked tags are visible to reader
- **Gen2 works fine on stacks of DVDs & racks of clothes**



Gen2 Read Range is Selectable

- **Read range is a function of antenna design!**
 - Far-field Gen2 tags can read at 10 meters
 - Near-field Gen2 tags can be restricted to less than a meter
- **With UHF Gen2 , you get what you need**
 - Short range, long range, or both!



**Satellite™ near-field PLUS
far field UHF Gen2 antenna**

Gen2 is Blazingly Fast

- Gen2 reads tags at rates exceeding 1000 tags/sec
 - No other RFID systems even come close

MultiReader - MACH1 Speedway Gen 2 RFID

File Edit Settings Graphs Help

INVENTORY RUN MODE

< Prev Run Next Run >

Impinj

Exit

Settings

☒ Use Reader1 ☐ Use Reader2

☐ Test Run Mode ☒ Color Age Tags

☒ Time Normalized ☐ Notification Trace

Test Run ID

Test Description

Inventory Statistics

Tags Total: 127

Tags Rdr1: 127

Tags Rdr2: 0

Tags Read: 127

Tot Reads: 3205101

Tot Rdr1: 3205101

Tot Rdr2: 0

Tags tx-t0: 2975.518

Test Time: 2965.518

Avg Rate: 1077.2

Curr Rate: 1077.2

Peak Rate: 1191.9

Inventory Operations

Test Time - ms

100000000

START Inventory

STOP Inventory Run

SAVE ALL Inv + Hist

REPEAT Inventory Run

CLEAR Inventory

	EPC VALUE	TOTCNT	RD1CNT	RD2CNT	RDRATE	TIMEt0	TIMEtx	Ttx-t0	RSSmx	NOC
124	3000214160C004000009255E	23985					5.487	2975.488	-97.3	
126	3000214160C0040000090C8A	24136					5.307	2975.307	-97.3	
115	3000214160C0040000090C97	25588					5.507	2975.507	-97.3	
109	3000214160C0040000092584	26894	26894	0	9.0	0.000	2975.287	2975.288	-97.3	
15	3000214160C0040000090C80	24472	24472	0	8.2	0.000	2975.457	2975.457	-97.3	
23	3000214160C004000009253D	25322	25322	0	8.5	0.000	2975.457	2975.457	-97.3	
114	3000214160C0040000091C42	24150	24150	0	8.1	0.000	2975.487	2975.487	-97.3	
82	3000214160C0040000091C9B	25901	25901	0	8.7	0.000	2975.447	2975.448	-97.3	
1	3000214160C0040000090CDD	24177	24177	0	8.1	0.010	2975.357	2975.348	-97.3	
11	3000214160C0040000091C39	25355	25355	0	8.5	0.010	2975.457	2975.447	-97.3	
111	3000214160C0040000090C01	24307	24307	0	8.2	0.010	2975.487	2975.477	-97.3	
21	3000214160C0040000090CC4	24825	24825	0	8.3	0.010	2975.347	2975.338	-97.3	
14	3000214160C0040000092534	24249	24249	0	8.1	0.010	2975.497	2975.487	-97.3	
84	3000214160C0040000091C41	24836	24836	0	8.3	0.010	2975.277	2975.267	-97.3	
59	3000214160C0040000090CCA	24621	24621	0	8.3	0.010	2975.517	2975.507	-97.3	
18	3000214160C0040000092559	26766	26766	0	9.0	0.010	2975.447	2975.437	-97.3	
7	3000214160C0040000090C80	26026	26026	0	8.7	0.010	2975.507	2975.497	-97.3	
117	3000214160C0040000090C8C	24208	24208	0	8.1	0.010	2975.517	2975.507	-97.3	
22	3000214160C0040000090CC6	25733	25733	0	8.6	0.010	2975.497	2975.487	-97.3	
12	3000214160C0040000090C99	24964	24964	0	8.4	0.010	2975.457	2975.447	-97.3	
119	3000214160C004000009254D	24890	24890	0	8.4	0.019	2975.327	2975.308	-97.3	
24	3000214160C0040000092531	24078	24078	0	8.1	0.020	2975.517	2975.497	-97.3	
85	3000214160C0040000090C79	24241	24241	0	8.1	0.020	2975.297	2975.277	-97.3	
127	3000214160C004000009259D	24115	24115	0	8.1	0.020	2975.517	2975.497	-97.3	
86	3000214160C0040000090CAE	24565	24565	0	8.3	0.020	2975.337	2975.317	-97.3	
62	3000214160C0040000092578	25217	25217	0	8.5	0.020	2975.467	2975.447	-97.3	
13	3000214160C004000009253B	26837	26837	0	9.0	0.020	2975.467	2975.447	-97.3	
5	3000214160C0040000090C8D	25077	25077	0	8.4	0.020	2975.487	2975.467	-97.3	
47	3000214160C0040000092556	25568	25568	0	8.6	0.020	2975.477	2975.457	-97.3	

Running Inventory Test 1 ...

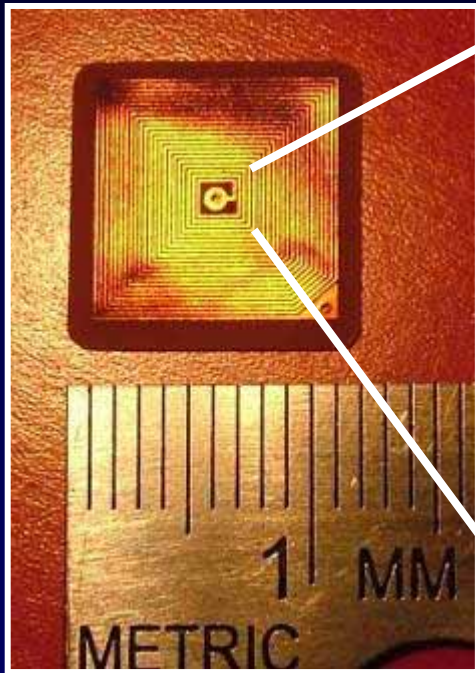
Reader1: Running

Reader2: Not Connected

07:33:38 PM

Key Gen2 Benefit: Low Tag Cost

HF Item-Level Inlay



~20 turns fine resolution etched copper, 2 layers, 12 mm diameter



Close-up showing high manufacturing complexity

UHF Item-Level Inlay



1 turn, 1 layer conductive ink, 9 mm diameter

UHF Gen2 Can Do It All

- **A single, open, worldwide standard**
 - A single infrastructure that can read items and pallets
 - Operation in the near-field, far-field, or both



An item-level UHF system

- Near-field coupling
- Confined 60cm read range
- Using the existing Gen2 protocol
 - Same Gen2 tag silicon
 - Same Gen2 readers
 - Simply change the antennas

Questions

